# CS 6515: Graduate Algorithms

Lecture 9: Linear Programming: Models and Geometry

Instructors: Sahil Singla and Gerandy Brito

Fall 2026

## 1 Announcements

1. Exam 1 will be graded by next week. We will normalize scores so that median is 86.
2. HW3 will be released soon.

## 2 Part 2: optimization as an algorithmic tool

In Part 1, we designed algorithms using divide and conquer, dynamic programming, and graph structure. Part 2 adds a different viewpoint: *describe the feasible solutions and the objective mathematically, then exploit the structure of the resulting optimization problem.*

| Block | Main questions |
|---|---|
| Linear programming (9–11) | How do we model a problem with linear constraints? How can we certify that a solution is optimal? |
| Convex optimization (12–13) | What survives beyond linear functions? How does gradient descent find an approximately optimal solution? |
| Approximation algorithms (14–16) | When exact discrete optimization is difficult, how can relaxations, rounding, and lower bounds give provably good solutions? |

**Connections to things we already know.** Maximum flow will be our first graph example of an LP. The knapsack FPTAS from Lecture 5 was already a preview of approximation algorithms. Later in Part 2, we will use LPs as relaxations of discrete problems. For convex optimization, the input and the objective are fixed and available throughout the computation.

## 3 What is a linear program?

A linear program (LP) has real-valued variables, a linear objective, and linear equality or non-strict inequality constraints. Throughout the LP lectures, our *canonical maximization form* is

$$\begin{array}{ll} \text{maximize} & c^\top x \\ \text{subject to} & Ax \le b, \\ & x \ge 0, \end{array} \qquad A \in \mathbb{R}^{m\times n}, \quad b \in \mathbb{R}^m, \quad c \in \mathbb{R}^n.$$

The inequalities are coordinatewise. The $j$-th row is $\sum_{i=1}^{n} A_{ji}x_i \le b_j$. For computational statements, the input coefficients are rational.

**Definition (feasible region).** The feasible region is the set of vectors satisfying *all* constraints:

$$P = \{x \in \mathbb{R}^n : Ax \le b,\ x \ge 0\}.$$

An optimal solution is a feasible vector with the best objective value.

### 3.1 Running example: making smoothies

We have 3 units of oranges, 10 of bananas, and 6 of strawberries. We can make fractional liters; the ingredients and revenue per liter are:

| Smoothie | Oranges | Bananas | Strawberries | Revenue |
|---|---|---|---|---|
| 1 | 1 | 2 | 0 | \$5 |
| 2 | 0 | 1 | 2 | \$4 |
| 3 | 1 | 1 | 1 | \$6 |

Let $x_i$ be the number of liters of smoothie $i$. The LP is

$$\begin{aligned} \text{maximize} \quad & 5x_1 + 4x_2 + 6x_3 \\ \text{subject to} \quad & x_1 + x_3 \le 3 \qquad \text{(oranges)}, \\ & 2x_1 + x_2 + x_3 \le 10 \qquad \text{(bananas)}, \\ & 2x_2 + x_3 \le 6 \qquad \text{(strawberries)}, \\ & x_1, x_2, x_3 \ge 0. \end{aligned} \tag{1}$$

The vector $(3, 3, 0)$ is feasible and earns 27. This proves OPT $\ge 27$, but *does not yet prove optimality.*

**Exercise.** Write the matrix $A$ and vectors $b, c$. What changes if smoothies can only be sold in whole-liter batches?

## 4 Modeling familiar problems and changing forms

### 4.1 Maximum flow is an LP

For a directed network, give each edge a variable $f_e$. As in Lecture 8, the value is net flow out of the source:

$$\begin{aligned} \text{maximize} \quad & \sum_{(s,v)\in E} f_{sv} - \sum_{(v,s)\in E} f_{vs} \\ \text{subject to} \quad & \sum_{(u,v)\in E} f_{uv} = \sum_{(v,w)\in E} f_{vw} \quad (v \notin \{s,t\}), \\ & 0 \le f_e \le c_e \quad (e \in E). \end{aligned}$$

There are $m = |E|$ variables and $\mathcal{O}(m+n)$ constraints. An LP formulation does not replace the specialized flow algorithms; it places flow in a broader optimization framework.

**Real variables versus integer variables.** The LP allows fractional flows. Lecture 8 proved that integer capacities guarantee an *integral optimal flow.* This is special structure: an arbitrary LP need not have an integral optimum.

### 4.2 Equivalent LP forms

| Feature | Conversion to canonical form |
|---|---|
| Minimize $c^\top x$ | Maximize $-c^\top x$, then negate the value. |
| $a^\top x \geq d$ | Write $-a^\top x \leq -d$. |
| $a^\top x = d$ | Write both $a^\top x \leq d$ and $-a^\top x \leq -d$. |
| Unrestricted variable $z$ | Substitute $z = z^+ - z^-$, with $z^+, z^- \geq 0$. |

These transformations increase the formulation size only polynomially. Another useful form is $Ax + s = b,\ x, s \geq 0$, obtained by adding *slack variables* $s = b - Ax$.

**What is not a plain LP?** Constraints such as $x_1 x_2 \leq 1$, an objective such as $x_1^2 + x_2^2$, or requirements $x_i \in \{0, 1\}$ are not part of an LP. We also do not allow strict inequalities: maximizing $x$ subject to $x < 1$ has supremum 1, but no maximizing point.

**Exercise.** Convert $\min(2u - v)$, subject to $u - v = 3$ and $u \geq 0$, into canonical form when $v$ is unrestricted.

## 5 The geometry of an LP

**Definition (convex set).** A set $P \subseteq \mathbb{R}^n$ is convex if, for every $x, y \in P$ and $\lambda \in [0, 1]$, the point $\lambda x + (1 - \lambda)y$ also belongs to $P$. Equivalently, the whole line segment between any two points of $P$ stays in $P$.

**Theorem (LP feasible regions are convex).** The feasible region of every LP is convex.

**Proof.** If $Ax \leq b$ and $Ay \leq b$, then

$$A(\lambda x + (1 - \lambda)y) = \lambda Ax + (1 - \lambda)Ay \leq \lambda b + (1 - \lambda)b = b.$$

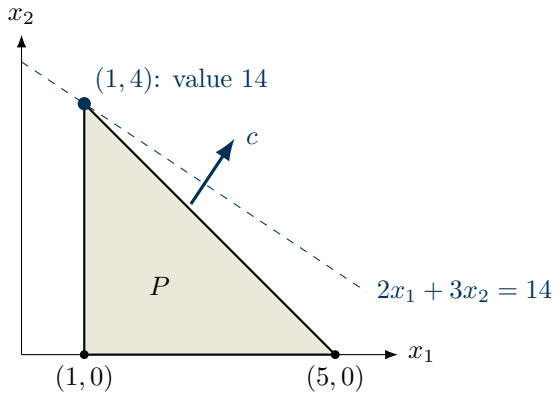
Nonnegativity is also preserved. Equalities are preserved by the same calculation.

### 5.1 A two-dimensional picture

Consider

$$\max\ 2x_1 + 3x_2 \quad \text{subject to} \quad x_1 + x_2 \leq 5, \quad x_1 \geq 1, \quad x_2 \geq 0.$$

Its feasible region is the triangle below. Lines of constant objective have equation $2x_1 + 3x_2 = q$; increasing $q$ moves them in direction $c = (2, 3)$.




The segment stays feasible. The best objective line touches a vertex.

If the objective were $x_1 + x_2$, an entire edge would be optimal. Thus an optimum need not be unique.

## 6 Vertices and the three possible outcomes

An LP can be:

- **infeasible:** no feasible point, e.g., $x \geq 2,\ x \leq 1$;
- **unbounded in objective:** feasible points have arbitrarily large objective, e.g., maximize $x$ subject to $x \geq 0$;
- **feasible with finite optimum:** then an LP attains its optimum.

An unbounded *feasible region* does not imply an unbounded objective: maximize $-x$ subject to $x \geq 0$.

**Definition (tight constraint and vertex).** An inequality $a^\top x \leq d$ is tight at $x$ if $a^\top x = d$. A feasible point of a polyhedron in $\mathbb{R}^n$ is a vertex (or basic feasible solution) if the normals of its tight constraints span $\mathbb{R}^n$: there are $n$ linearly independent tight constraints.

Nonnegativity constraints count: write $x_i \geq 0$ as $-x_i \leq 0$. A vertex can have more than $n$ tight constraints; independence matters.

**Theorem (a basic optimal solution).** If an LP in canonical form $Ax \leq b,\ x \geq 0$ is feasible with finite optimal value, it has an optimal solution at a vertex.

**Geometric intuition.** If fewer than $n$ independent constraints are tight, there is a nonzero direction parallel to all of them. A sufficiently small move in either direction remains feasible. At an optimum, this direction cannot improve the objective, so the objective is constant along it. Move until another independent constraint becomes tight, and repeat. Nonnegativity prevents a whole line from remaining feasible in both directions forever.

**A useful caveat.** Without the canonical-form assumption, a polyhedron can have no vertices. For example, the half-plane $\{(u, v) : v \geq 0\}$ has none, although $\max(-v)$ has finite optimum 0. Converting to canonical form restores the vertex statement in the new variables.

**Exercise.** For the two-dimensional triangle, identify two independent tight constraints at each vertex. At $(3, 3, 0)$ in the smoothie LP, identify three independent tight constraints.

# 7 Why LPs are algorithmically useful

**Theorem (polynomial-time solvability, used as a black box).** For rational input, an LP can be solved in time polynomial in its total binary encoding length: report infeasibility or unboundedness, or return an exact rational optimum when the optimal value is finite.

The ellipsoid method and interior-point methods underlie polynomial-time LP algorithms. We will use LP solving as a tool, not develop these algorithms.

## 7.1 Why can the answer be written with polynomially many bits?

At a vertex, select $n$ independent tight constraints, including nonnegativity constraints if needed. They give a nonsingular system

$$Bx = d.$$

By Cramer's rule, $x_i = \det(B_i)/\det(B)$, where $B_i$ is obtained by replacing column $i$ of $B$ with $d$. After clearing rational denominators, determinants of integer matrices have polynomial encoding length: each is a sum of $n!$ products of $n$ entries. Hence a basic optimal solution has polynomial bit length.

**This is not an enumeration algorithm.** There may be exponentially many choices of tight constraints. The existence of a short answer does not imply that testing every candidate vertex is efficient.

**Connection to our knapsack discussion.** Polynomial in *input bit length* differs from polynomial in the numerical magnitudes. If a capacity is $B = 2^r$, its encoding uses $r + 1$ bits, whereas a DP taking $\mathcal{O}(nB)$ time is pseudo-polynomial. An LP algorithm's polynomial guarantee is in the encoding length; do not read this theorem as saying the running time depends only on the numbers of variables and constraints.

**Exercise.** An LP in $n$ variables has box constraints $0 \le x_i \le 1$ and only $k$ other linear inequalities. Show that any vertex has at most $k$ coordinates strictly between 0 and 1.

# 8 How do we prove a proposed solution is optimal?

Return to the smoothie LP.

$$\begin{aligned} \text{maximize} \quad & 5x_1 + 4x_2 + 6x_3 \\ \text{subject to} \quad & x_1 + x_3 \le 3 \qquad \text{(oranges)}, \\ & 2x_1 + x_2 + x_3 \le 10 \qquad \text{(bananas)}, \\ & 2x_2 + x_3 \le 6 \qquad \text{(strawberries)}, \\ & x_1, x_2, x_3 \ge 0. \end{aligned} \tag{2}$$

The feasible solution

$$x = (3, 3, 0) \qquad \text{has value} \qquad 5 \cdot 3 + 4 \cdot 3 = 27.$$

To prove it is optimal, we need an upper bound of 27 for *every* feasible solution.

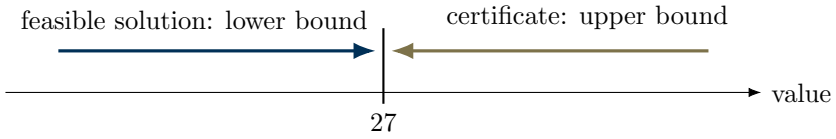
Take five times the orange constraint and two times the strawberry constraint:

$$5(x_1 + x_3) + 2(2x_2 + x_3) \leq 5 \cdot 3 + 2 \cdot 6 = 27.$$

The left side is $5x_1 + 4x_2 + 7x_3$. Since $x_3 \geq 0$,

$$\boxed{5x_1 + 4x_2 + 6x_3 \leq 5x_1 + 4x_2 + 7x_3 \leq 27.}$$

Equivalently, include one copy of the constraint $-x_3 \leq 0$ to make the left side exactly the objective.




**The next question.** We found nonnegative weights $(5, 0, 2)$ on the three resource constraints. Can we systematically search for the *best* such upper bound? That search is itself an LP: the **dual**.

**Takeaways.** An LP separates the model from the algorithm. Convex geometry explains its structure, and a short certificate can prove optimality without requiring us to rerun the optimization algorithm.